

# Deep and Morphological tenselessness: Evidence from Kanien'kéha conditionals

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## 1 Background

Kanien'kéha event verbs appear in one of three aspectual forms.

|                            |                          |                        |
|----------------------------|--------------------------|------------------------|
| (1) a. <b>Perfective</b>   | b. <b>Imperfective</b>   | c. <b>Perfect</b>      |
| wa'kenóhare'               | ke'nóhares               | wakenóhare             |
| wa'-ke-nohare-'            | ke-nohare-s              | wake-nohare            |
| FACT-1SGA-wash- <b>PFV</b> | 1SGA-wash- <b>IPFV</b>   | 1SGP-wash- <b>PERF</b> |
| 'I washed it'              | 'I wash it (habitually)' | 'I have washed it'     |

There are two morphological asymmetries that differentiate between PERFECTIVE aspect and the two non-perfective (IMPERFECTIVE and PERFECT) aspects:

- ① Non-perfective aspects are optionally followed by tense suffixes; perfective aspect never is (2)
- ② Perfective aspects require one of three modal/mood prefixes (FACTUAL, FUTURE, OR OPTATIVE), whereas non-perfective aspects require a "modalizer" suffix to co-occur with modal/mood prefixes (3)<sup>1</sup>

|                                           |                                          |
|-------------------------------------------|------------------------------------------|
| (2) <b>Overt tense suffixes asymmetry</b> | (3) <b>Modal/mood prefixes asymmetry</b> |
| a. katá:wenskwe'                          | a. enkatá:wen'                           |
| k-atawen-s- <b>kwe'</b>                   | en-k-atawen-'                            |
| 1SGA-swim-IPFV- <b>FOR.PST</b>            | FUT-1SGA-swim-PFV                        |
| 'I used to swim'                          | 'I will swim'                            |
| b. *wa'katá:wen'kwe'                      | b. enkatawénhseke'                       |
| (wa'-)k-atawen-'- <b>kwe'</b>             | en-k-atawen-hs-*( <b>eke'</b> )          |
| FACT-1SGA-swim-PFV- <b>FOR.PST</b>        | FUT-1SGA-swim-IPFV- <b>MODZ</b>          |
| Intended: 'I swam'                        | 'I will be swimming'                     |

<sup>1</sup>With non-perfective aspects, only the future and optative prefixes are possible. The modalizer is sometimes null with the perfect in Kanien'kéha but always overt in related languages.

Previously, Baker and Travis (1997, 1998) have accounted for these morphological asymmetries by the following assumption:

(4) **Tenselessness of perfectives**

Non-perfective aspect introduces a time argument, whereas perfective aspect verbs do not.

Empirical support for this assumption:

① **Non-perfectives are tensed (introduce a time argument)**

Gatchalian (2025) argues (from cessation implicatures and identity under ellipsis) that the non-perfective aspects are always tensed; when they do not have the optional tense suffix, they are *covertly tensed*.

⇒ This account raises the question of whether perfective forms (for which there are no overt tense suffixes) are also covertly tensed.

② **Perfectives are not tensed (make no reference to times)**

This poster presents data from counterfactual conditionals as a diagnostic for the lack of a temporal semantics with the perfective aspect. (See similar conclusions from Yucatec Maya by Bohnemeyer 2019)

**Takeaway**

**Empirically,**

- ① Counterfactuality in Kanien'kéha is restricted to imperfective and perfect aspect verbs
- ② Perfective aspect both lacks overt tense morphology and is unable to express counterfactuals.

**Theoretically,**

- ⇒ We link the differential behaviour of the aspects with respect to counterfactuality to Baker and Travis's (1997; 1998) differential presence of *tense*, based on aspect (4)
- More broadly, counterfactuality can be used as a diagnostic for tenselessness for languages with the correct morphosyntactic profile (Bohnemeyer 2019)

## 2 Central generalizations

### 2.1 Generalization 1

**Generalization 1:** The optative prefix is necessary for counterfactual (CF) interpretations.

- Modal-less (5a) and future-marked (6a) verbs are ruled out in CF contexts

(5) **Context:** *I'm sitting in my office, tired of work.*

- a. #Sénha wakatshennón:ni' tóka' tsi kaniataratátie' tkatá:wens  
 senha wak-atshennoni' toka tsi kaniatar-atatie' t-k-atawen-s  
 more 1SGP-be.happy.STAT if that river-along CIS-1SGA-swim-IPFV  
 Int.: 'If I were swimming by the beach, I would be happier.' (IPFV)
- b. Sénha aonkwatshennónníhake' tóka' tsi kaniataratátie' aontakatawénsheke'.  
 senha aon-wak-atshennoni-hake' toka tsi kaniatar-atatie' aon-ta-k-atawen-s-heke'  
 more OPT-1SGP-be.happy.PERF-MODZ if that river-along OPT-CIS-1SGA-swim-IPFV-MODZ  
 'If I were swimming by the beach, I would be happier.' (OPT-IPFV)

(6) **Context:** *I am not rich, and I do not have a big house.*

- a. #tóka' enwakkwatshéheke', kanonhsowá:nen enwakenonhsó:take'  
 toka en-wak-kwatshe-heke' ka-nonhs-owanen en-wake-nonhs-ot-a-ke'  
 if FUT-1SGP-be.rich.PERF-MODZ 1SGA-house-be.big.PERF FUT-1SGP-house-stand-PERF-MODZ  
 Int.: 'If I were rich, I would have a big house.' (FUT-PERF)
- b. tóka' aonkkwatshéheke', kanonhsowá:nen aonkenonhsó:take'  
 toka a-wak-kwatshe-heke' ka-nonhs-owanen a-wake-nonhs-ot-a-ke'  
 if OPT-1SGP-be.rich.PERF-MODZ 1SGA-house-be.big.PERF OPT-1SGP-house-stand-PERF-MODZ  
 'If I were rich, I would have a big house.' (OPT-PERF)

**Generalization 2a:** The optative prefix is not sufficient for counterfactual interpretations

- Optative marked *perfective* aspect verbs cannot receive counterfactual interpretations (7a)
- Instead, the perfect (7b) or the imperfective (8) are required in counterfactual contexts, depending on the aspectual interpretation.

(7) **Context:** *There is a gathering next week, and there will be dance performances. My late mother was a very good dancer, and people loved to watch her.*

- a. #Tóka' isten'kénha tsiahià:khsera tá:we' taienónniahkwe', akwé:kon  
 toka isten'-kenha tsiahiakhsera tawe' t-a-ie-nonniahkw-e' akwekon  
 if my.mother-DECESS week coming DUPL-OPT-FLA-dance-PFV all  
 nón:kwe aiakaon'eskwani:'on/aiakaon'éskwen'.  
 ne=onkwe a-iaka-on'esw-a-ni-'on/a-iaka-on'esw-en  
 NE=people OPT-FL.O-enjoy-BEN-JR-PERF/OPT-FL.O-enjoy-BEN-PFV  
 Int.: 'If my late mother was dancing next week, everyone would have liked it.' (OPT-PFV)
- b. Tóka' isten'kénha tsiahià:khsera tá:we' taiakononniáhkwen, akwé:kon  
 toka isten'-kenha tsiahiakhsera tawe' t-a-iako-nonniahkw-en akwekon  
 if my.mother-DECESS week coming DUPL-OPT-FLP-dance-STAT all  
 nón:kwe aiakaon'eskwani:'on.  
 ne=onkwe a-iaka-on'eswa-ni-'on  
 NE=people OPT-FL.O-enjoy-BEN-PERF  
 'If my late mother was dancing next week, everyone would have liked it.' (OPT-PERF)

- (8) **Context:** *Kor is studying to be a writer. Wari wants to marry a cook.*

Kó:r enhahiatónseke’

Kor en-ha-hiaton-s-eke’

Kor FUT-MSGA-write-IPFV-MODZ

Toka’ ahakhonniseke’

ne Kór, Onwári Kó:r ahotiniá:kon

toka a-ha-kh-onni-s-eke’

ne Kor Onwari Kor a-hoti-niakon

if OPT-MSGA-food-make-IPFV-MODZ NE Kor Onwari Kor OPT-MNSGP-marry.PERF

‘Kor will be a writer. If Kor would have been a cook instead, Wari would want to marry him!’ (OPT-IPFV)

**Generalization 2b:** The optative with non-perfective aspects have free temporal reference

- (9) a. *Past counterfactual*

Aonkenonhwaktani:’on tóka’ thí:ken sewahió:wane’ aonkwahíá:kon.

a-wake-nonhwakt-ani’-on toka thiken sewahiwane’ a-wak-ahi-a-k-on’

OPT-ISGP-be.sore-BEN-PERF if this apple OPT-ISGP-fruit-JR-eat-PERF

Ioiánere’ tsi iah tewakahíá:kon!

io-ianer-e’ tsi iah te-wak-ahi-a-k-on

NP-be.good-PERF that NEG NEG-ISGP-fruit-JR-eat-PERF

‘If I had eaten this apple, I would have gotten sick. Good thing I didn’t.’ (OPT-PERF)

- b. *Present counterfactual*

Tóka’ aonkkwatshéheke’, kanonhsowá:nen aonkenonhsó:take’

toka a-wak-kwatshe-heke’ ka-nonhs-owanen a-wake-nonhs-ot-a-ke’

if OPT-ISGP-be.rich.PERF-MODZ ISGA-house-be.big.PERF OPT-ISGP-house-stand-PERF-MODZ

‘If I were rich, I would have a big house.’ (OPT-PERF)

- c. *Future counterfactual*

Tóka’ isten’kénha tsiahíà:khsera tá:we’ taiakononniáhkwen, akwé:kon

toka isten’-kenha tsiahíakhsera tawe’ t-a-iako-nonniáhkwen akwekon

if my.mother-DECESS week coming DUPL-OPT-FL.P-dance-PERF all

nón:kwe aiakaon’eskwani:’on.

ne=onkwe a-iaka-on’eswa-ni-’on

NE=people OPT-FL.O-enjoy-BEN-PERF

‘If my late mother was dancing next week, everyone would have liked it.’ (OPT-PERF)

### CF morphology

Counterfactuality is signalled by the OPTATIVE and a NON-PERFECTIVE aspect (with the modalizer):

- (10) a. OPT-V-IPFV/PERF-CONT  $\rightsquigarrow$  CF

- b. OPT-V-PFV  $\nrightarrow$  CF

**Generalization 3a:** OPT+PFV may give rise to a possible future interpretation

(11) **Context:** *There is a gathering next week, and there will be dance performances. My mother is a very good dancer, and people love to watch her.*

- a. Tóka' istén'ha tsiahià:khsera tá:we' tenienónnniahkwe', akwé:kon nón:kwe  
 toka isten'ha tsiahiakhsera tawe' t-en-ie-nonniahkw-e' akwekon ne=onkwe  
 if my.mother week coming DUPL-FUT-FL.A-dance-PFV all NE=people  
 aiakaon'éskwen'.  
 en-iaka-on'esw-en  
 FUT-FL.O-enjoy-BEN-PFV  
 'If my mother dances next week, everyone will like it.' (FUT-PFV)
- b. Tóka' istén'ha tsiahià:khsera tá:we' taienónnniahkwe', akwé:kon nón:kwe  
 toka isten'ha tsiahiakhsera tawe' t-a-ie-nonniahkw-e' akwekon ne=onkwe  
 if my.mother week coming DUPL-OPT-FL.A-dance-PFV all NE=people  
 aiakaon'éskwen'.  
 a-iaka-on'esw-en  
 OPT-FL.O-enjoy-BEN-PFV  
 'If my mother danced next week, everyone would like it.' (OPT-PFV, ~“future less vivid”)
- c. #Tóka' istén'ha tsiahià:khsera tá:we' taiakonónnniahkwén, akwé:kon nón:kwe  
 toka isten'ha tsiahiakhsera tawe' t-a-ie-nonniahkw-e' akwekon ne=onkwe  
 if my.mother week coming DUPL-OPT-FL.P-dance-PERF all NE=people  
 aiakaon'éskwani'on.  
 a-iaka-on'esw-ani-on  
 OPT-FL.P-enjoy-BEN-PERF  
 Int.: 'If my mother danced next week, everyone would like it.' (OPT-PERF)

**Generalization 3a:** opt+pfv cannot get past or present interpretations

- Present interpretations are ruled out independently – perfective aspect cannot generally give rise to present readings outside of performative contexts (and with some perception verbs, yet to be tested in counterfactual contexts)
- Past readings require the perfect (as seen above), repeated in (12)

(12) **Context:** *I see a rotten apple on the counter, and throw it away.*

- a. #Aonkenonhwaktani'on tóka' thí:ken sewahió:wane' akà:iake'.  
 a-wake-nonhwakt-ani'-on toka thiken sewahiwane' a-k-ahi-a-k-e'  
 OPT-ISGP-be.sore-BEN-STAT if this apple OPT-ISGA-fruit-JR-eat-PUNC  
 Ioiánere' tsi iah tewakahiá:kon!  
 io-ianer-e' tsi iah te-wak-ahi-a-k-on  
 NP-be.good-STAT that NEG NEG-ISGP-fruit-JR-eat-STAT  
 Int.: 'If I had eaten this apple, I would have gotten sick. Good thing I didn't.' (OPT-PFV)

- b. Aonkenonhwaktani:’on      tóka’    thí:ken    sewahió:wane’    aonkwahíá:kon.  
a-wake-nonhwakt-ani’-on    toka    thiken    sewahiwane’    a-wak-ahi-a-k-on’  
OPT-1SGP-be.sore-BEN-STAT    if    this    apple    OPT-1SGP-fruit-JR-eat-STAT  
loíanere’      tsi    iah    tewakahíá:kon!  
io-ianer-e’      tsi    iah    te-wak-ahi-a-k-on  
NP-be.good-STAT    that    NEG    NEG-1SGP-fruit-JR-eat-STAT  
‘If I had eaten this apple, I would have gotten sick. Good thing I didn’t.’      (OPT-PERF)

| meaning      | OPT-{IPFV/PERF} | OPT-PFV |
|--------------|-----------------|---------|
| past CF      | ✓ (9a)          | #       |
| pres CF      | ✓ (9b)          | #       |
| fut CF       | ✓ (9c)          | #       |
| possible fut | # (11c)         | ✓ (11b) |

### 3 Towards an Account

#### 3.1 Preliminaries

**Tense is implicated in counterfactual interpretations in some languages**

- Cross-linguistically, counterfactual markers (X-markers; von Fintel and Iatridou 2023) tend to be temporal morphemes, though they appear to be stripped of their temporal meanings
  - In languages where X-markers are temporal morphemes, it is overwhelmingly the exponent for PAST which is used as the X-marker (but see Ferreira and Müller 2019 on Karitiana’s use of NFUT)
  - Accounts aiming to make sense of this tendency typically focus on languages with obligatory tense inflection (though see Zimmermann 2021 on Hausa).
- ⇒ Whether one treats the temporal morphology as semantically temporal (Ippolito 2013) or as remoteness marker (Iatridou 2000), the *presence of tense* is still implicated

**Tense is the X-marker in Kanien’kéha**

- Kanien’kéha counterfactuals are **doubly-marked**: they require both a non-perfective aspect and the Optative prefix. Neither alone is sufficient for a counterfactual interpretation.
- Cross-linguistically, temporal X-markers may be PAST tense or IMPERFECTIVE aspect. Languages that appear to be double-marked by temporal morphemes have been argued to be due to morphological conflation of the tense and aspect features (Bjorkman and Halpert 2017)
- However, languages with a *modal/mood component* to their X-markers may still have restrictions on the temporal morphemes that may be embedded: the Russian subjunctive *by* still requires past tense (Bjorkman and Halpert 2017) and Karitiana’s counterfactual marker *jĩ* still requires nonfuture tense (Ferreira and Müller 2019)

- ⇒ If the X-marker in Kanien'kéha is aspect, we miss the generalization that a single property shared by these aspects is getting us CF: the (possibly past) temporal reference time that comes with IPFV/PERF.
- Note also that the interpretation of the embedded aspect is never fake – it always receives the value typically associated with it (perfect, ongoing, or habitual)
- ⇒ To account for the aspectual restriction above, we will argue: ① Kanien'kéha is a temporal X-marking language, more specifically it uses TENSE, and ② it is the *absence of tense wholesale* which prevents perfective aspect verbs from expressing counterfactuality

### 3.2 Proposal: Perfectives are tenseless

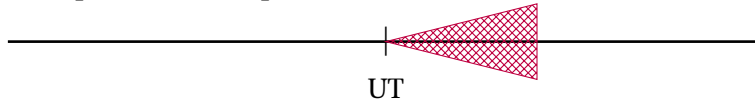
All theories of the role of tense in counterfactual conditionals assume that the tense is interpreted, either with its usual temporal meaning (Arregui 2007, 2009; Ippolito 2013) or a more general exclusionary meaning (Iatridou 2000; von Prince 2019). To see this, let us be relatively concrete and assume the following three-argument meaning of the conditional modal (adapted from Arregui 2009; Ippolito 2013; Ippolito and Mirrazi 2025):

$$(13) \quad \llbracket \text{OP}_{\text{CF}} \rrbracket^c(p)(q)(t) = 1 \text{ in } w_c \text{ at } t_c \text{ iff } \exists t : t < t_c [\forall w : w \in \text{HIST}_{w_c, t} [p(w) = 1] \rightarrow [q(w) = 1]]$$

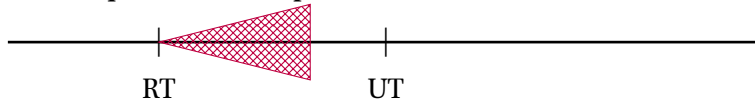
This operator takes two propositions and a time argument  $t$ , evaluated at  $w$ . It states that all p-worlds accessible from  $w$  at  $t$  are also q-worlds.

- Assuming a branching-times model, counterfactual worlds are those which have diverged from the evaluation world at some point in the past (von Prince 2019)
- Thus, to access counterfactual worlds, we relativize the time of the accessibility function to some point in the past, and consider those worlds developing divergently from  $w_c$  from that point

- (14) a. OPT quantifies over possible continuations from the UT



- b. OPT+T quantifies over possible continuations from the RT



The proposal under consideration, then, is that in Kanien'kéha, only imperfective and perfect aspects provide the semantic resources to access counterfactual worlds – and that these resources are *temporal*.

- (15) a.  $[\text{Mod}_{\text{OPT}} [\text{Asp}_{\text{PFV}} [\text{VP}]]] \rightsquigarrow$  modal base: worlds accessible from UT
- b.  $[\text{Mod}_{\text{OPT}} [\text{T}_{\text{CONT}} [\text{Asp}_{\text{IPFV/PERF}} [\text{VP}]]]] \rightsquigarrow$  modal base: worlds accessible from  $t < \text{UT}$

⇒ Note, however, that it is the morphological and semantic resources of the *propositions embedded by the modal* that want to co-vary with the time argument of the modal itself in (13).

### Thus, there is a morphological linking problem

- either the time argument of the modal determines the form of the embedded predicates, or the tense of the embedded predicates must determine the value of the time argument.
- ⇒ Independent of this problem (which is faced by all accounts assuming an LF like (13)), if (past) tense is a crucial ingredient for counterfactuality, lacking *any* tense should rule out counterfactuality

### Lack of tense means obligatorily anchoring the modal/temporal reference to the utterance time

- Perfective aspect, by hypothesis, lack tense and therefore cannot refer to an shiftable reference time
  - That is, the temporal location of events of perfective aspect verbs are always interpreted relative to the utterance time.
- ⇒ For conditionals, the time of the optative marked verb is obligatorily future *relative to UT*

### Tense means we can shift the modal/temporal reference to some other time

- With non-perfective aspects, tense permits a reference time shift.
- ⇒ The time of the optative marked verb is future relative to some (possibly past) reference time, thus permitting access to counterfactual worlds.

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